

Lachlan James Harnett

lachlanjharnett@gmail.com | <https://rnett.au/>

Summary

Research software engineer at CSIRO, delivering backend software, cloud and HPC infrastructure, and automated research workflows across a range of research fields. I turn what researchers and business stakeholders need into well-defined, reliable solutions, and I'm passionate about technology that has a positive impact on society.

Experience

Research Software Engineer - CSIRO, Brisbane

Aug 2024 – Present

Scientific Computing group. Working with multidisciplinary research and product teams to design and deliver scalable solutions on high performance computing, cloud and containerised environments.

- Automated client onboarding for NatureIQ, CSIRO's biodiversity and sustainability SaaS platform, with AWS Lambda, Step Functions and Smithy, cutting dozens of manual steps to one. Also built its admin application and monitoring. Individually recognised with a CSIRO award and reward in 2025.
- Integrated a land-carbon science model into NatureIQ through a Python AWS Lambda interface, and built an R utility generating synthetic geospatial test data.
- Took ownership of Sciansa, an AI-assisted scientific analysis platform, at short notice. Rebuilt CI/CD on GitHub Actions with automated tests, and added a tools catalogue, versioning, live run updates and an agent registry.
- Refactored and recompiled CSIRO's WAVES model from Windows for the Petrichor HPC cluster, optimised it for HPC, and implemented a shuffled complex evolution optimiser to effectively parameterise it for TERN IV (Fortran, Python, Bash, Slurm).
- Built R and Shiny governance tooling for CSIRO's Posit Connect platform: daily scanners for misconfigured apps and invalid permissions, an account management app and load balancer dashboards.
- Built a web app for the enterprise Bitbucket to GitHub migration that automated ownership tickets for thousands of repositories.
- Learned ParaView under time pressure to animate compressed air energy storage data for external stakeholders.
- Presented on NatureIQ's architecture and on R and Shiny at internal seminars, and contribute to the Scientific Computing outreach working group.

Data Analyst, Graduate Program - CSIRO, Brisbane

Feb 2024 – Aug 2024

- Built an R Shiny application and automated data pipeline for the Marine Language Toolkit, used in two research papers.

Academic Tutor – Union College, Brisbane

March 2023 – November 2024

Tutored Computer Science, Software Engineering and Mathematics courses at UQ and QUT.

- Designed workshops to enhance the coding abilities of students from a variety of academic backgrounds.

Education

Queensland Certificate of Education

St. Augustine's College, Cairns

Graduated 2018 with OP 2 (ATAR 98.25)

Bachelor of Computer Science

University of Queensland, St. Lucia

Graduated 2023 with GPA 6.1

Skills

Certifications and Training

AI Industry Training & Hackathon – UNSW Fintech AI Innovation Consortium (2026)

Architecting on AWS – Lumify Work

CUDA Programming and HPC Storage Optimisation workshops – eResearch Australasia

Foundational Cybersecurity – Google

Introduction to AI – NSW TAFE

Technical Overview

Languages and Frameworks:

Python, R, TypeScript, JavaScript, C, C++, C#, SQL, MATLAB, React, Shiny

AI Engineering:

AI-assisted SDLC, loop engineering, spec-driven development with GitHub Spec Kit, building MCP servers and agent skills

Cloud Platforms:

AWS (Lambda, Step Functions), GCP, HPC clusters

Tools:

Docker, Git, GitHub Actions, Linux, Dask, ParaView, QGIS, Postgres, DynamoDB